

Name:

Unique ID:

Math 221D Quiz 1 (September 4, 2026)

1. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, and suppose that $\mathbf{x}, \mathbf{y} \in \text{Span}(\mathbf{u}, \mathbf{v})$. Prove that $\mathbf{x} - 5\mathbf{y} \in \text{Span}(\mathbf{u}, \mathbf{v})$ as well.
2. Prove that every vector $\mathbf{x} \in \mathbb{R}^3$ which satisfies $x_1 - 2x_2 + x_3 = 0$ is a linear combination of $\mathbf{u} = (1, 0, -1)$ and $\mathbf{v} = (1, 1, 1)$.